Parameter	Interpretation	Equations	Simulation Values	
division	division constant		0.02	
k_on	Activation constant	(drug_delta + 0.2*D_current)/(kd+D_current)		
k_off	Inactivation constant		100	
ktransc	Transcription constant		100	
ktransl	Translation constant		70	
mrna_decay	mRNA degradation constant		0.1	
prot_degrad	Protein degradation constant		0.5	
drug_delta	Maximum death rate drug can achieve		0.01-1	
K	Protein level to overcome the effects of the drug		0.001	
D_current	Current dosage		1, 10, 50, 100, 250, 500, 1000	
death	Drug-dependent death rate	drug_delta * ((D_current^n)/ (D_current^n + (K * prot) ^n))		
n	Hill coefficient resembling the strength of the dose-reponse		1, 1.5, 2	
x	Array of gene states			
RNA	Array of mRNA			
prot	Array of proteins			
active	Activation rate	k_on * (1-x)		
off	Inactivation rate	k_off * x		
transc	Transcription rate	ktransc * x		
transl	Translation rate	ktransl * RNA		
mrna_death	mRNA degradation rate	mrna_decay * RNA		
prot_death	Protein degradation rate	prot_degrad * prot		
div	Division rate	division * ones(1, N)		
N	Number of cells		50-100	
tmax	Maximum lifetime of a cell		50	
kd	induction midpoint dose where 50% of cells are killed			